

Design Report Concept

1. Application

Procreate (Category: Creation), a professional digital illustration and painting application.

2. User Tasks and Goals

a. Immersive Drawing Experience

Goal: Create freely on an expandable canvas without being restricted by screen size.

Task: Use controllers or hand gestures to zoom, rotate, and move the canvas in space.

b. Layered Drawing

Goal: Manage and refine complex artworks through separate editable layers.

Task: Switch between layers, show/hide them, and edit each individually for detailed adjustments.

c. Real-world Preview

Goal: Place and view the artwork in realistic environments for evaluation and presentation.

Task: Adjust scale and position to preview artwork on billboards, posters, products, or street walls.

d. Projection Assistance

Goal: Simulate projecting a sketch or artwork onto real-world surfaces such as walls or tables to replace physical projectors.

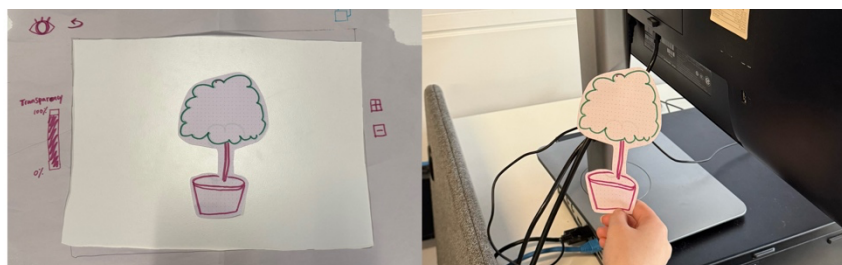
Task: Adjust transparency and scale through controllers, align the projection to the surface, and use it as a guide for large-scale real-world painting or mural work.

3. Idea development process

a. Adapt Procreate for XR to allow drawing in a large, freeform canvas and added multi-layer functionality for better control over complex artworks (Initial idea)



b. Added real-world preview to help artists evaluate and present designs in realistic locations (Based on a review of existing products)



c. Introduced projection assistance to support real-world mural and wall art creation without physical projectors (Based on week 2 studio lo-fi prototype test feedback)

4. Define the concept of Procreate in XR

Procreate XR will provide a fully spatial experience for digital art creation in a mixed reality environment, merging the physical environment with a virtual canvas. Users can create on an infinite

canvas, take advantage of XR's capabilities for more intuitive and efficient layer management, use virtual projection to assist real-world painting, and place their artwork directly into real-world contexts for preview and presentation. Procreate XR will support the following interactions:

- a. **Free Canvas**
Adjust and navigate the canvas using pinch, drag, and rotate gestures (or controllers).
- b. **Floating Layer Panel**
A floating menu that allows users to select, reorder, and toggle the visibility of layers.
- c. **Real-world Preview Mode**
Choose from preset scenes (billboards, posters, street walls, product packaging) or scan the real environment to anchor the artwork onto physical objects.



- d. **Projection Mode**
Overlay a semi-transparent version of the artwork onto any real-world surface, with adjustable scale, position, and opacity.



5. Initial testing plan

- a. Interactions and features to test
 - Canvas zoom/rotate/move gestures (or controllers).
 - Layer management (switching, editing, hiding).
 - Projection mode's accuracy, opacity control, and usability for real-world painting.
 - Real-world preview placement accuracy and realism.
- b. Assumptions
 - Users can intuitively navigate a spatial canvas with gestures or controllers.
 - Projection assistance will be more practical and precise than traditional projectors.
 - Real-world preview will improve creative decision-making and client presentations.
- c. Data collection and validation need
 - Record task completion time and error rates to validate the functionality is easy to use and supports efficient task performance.
 - Record usability testing results (e.g. Think aloud) to validate whether the interactions are smooth, intuitive, and align with user expectations.
 - Collect qualitative user feedback on workflow and overall satisfaction to identify opportunities for future improvement.